Muhammad Talha Shafi

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Education

National University of Computer and Emerging Sciences (FAST-NUCES) Islamabad, Pakistan
Bachelor of Science in Computer Science — CGPA: 3.56/4.00 *Aug 2023 – Jul 2027 (Expected)*

- **Dean's List of Honors** in four semesters: Fall 2023, Fall 2024, Fall 2025, Spring 2026
- **Relevant Coursework:** Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Machine Learning, Artificial Intelligence, Natural Language Processing, Digital Image Processing, High Performance Computing

Technical Skills

Languages: TypeScript/JavaScript, Python, C/C++, Java, SQL
Backend: Node.js, Express.js, FastAPI, REST API design, WebSockets, Server-Sent Events, Job Queues (Bull), JWT Auth, RBAC, Rate Limiting
Frontend & Mobile: React, Next.js, Tailwind CSS, HTML/CSS, Flutter
Databases: PostgreSQL (row-level security, indexing, transactions), MongoDB, MySQL, Redis, Supabase
DevOps & Cloud: Docker, GitHub Actions (CI/CD), Azure, Linux, Bash, Git, Sentry, Vercel, VPS deployment
Machine Learning & Vision: PyTorch, Pose Estimation (YOLO-Pose, ViTPose++), Temporal CNNs, OpenCV, OCR, NumPy, Pandas
LLM & Agentic Systems: LangChain, LangGraph, Multi-Agent Orchestration, RAG, LLM APIs (Claude, Gemini, OpenAI), n8n
Testing: pytest, Vitest, Playwright

Experience

Agent Evals Engineer Jun 2026 – Aug 2026
Horizon (Freelance, Part-time) *Remote*

- Engineered **20+ sandboxed Docker environments** to benchmark LLMs on long-horizon agentic tasks across software engineering, systems administration, and data science
- Designed deterministic **pytest grading suites** with anti-cheating checks; calibrated task difficulty to strict statistical bands (pass rate 0.1–0.4, average trajectory $\geq$15 turns)
- Diagnosed agent failure modes (tool-call cascades, context saturation, hallucination under timeout pressure); findings fed directly into task calibration

Software Engineer Mar 2026 – Jun 2026
Beaj Education (Part-time) *Remote*

- Integrated the **Meta Messaging API** into a production **Node.js/PostgreSQL** backend; shipped interactive URL messages and 2 new bot activity flows with Redis caching and Sentry monitoring
- Resolved an **Azure Blob Storage memory leak** and refactored 4 large modules, reducing coupling and improving build reliability across staging and production
- Built a facilitator dashboard unifying attendance, joining status, performance metrics, and stuck-student detection, replacing several manual spreadsheets

Co-Founder & Technical Lead Jul 2025 – Jun 2026
Meri Kahani Society, FAST-NUCES *Islamabad, Pakistan*

- Co-founded the society, now one of the largest at FAST-NUCES Islamabad; led its **10+ member** Web Development and Automation team and launched the official website
- Automated hiring and onboarding with **n8n pipelines**: outreach campaigns, email sequences, and member-directory sync through webhook-driven workflows and custom nodes

Lab Demonstrator, Digital Logic Design Jan 2025 – May 2025
FAST-NUCES *Islamabad, Pakistan*

- Guided **50+ students** through circuit design across **45+ hours** of lab instruction; ran evaluations and practical demos

Teaching Assistant, Programming Fundamentals Aug 2024 – Jan 2025
FAST-NUCES *Islamabad, Pakistan*

- Created assignments and a semester project for **6 CS sections (200+ students)**; ran weekly problem-solving sessions

Full-Stack Projects

Baton: Lead-Response CRM | *React, TypeScript, Supabase (PostgreSQL), Deno, Stripe* | Live — Oct 2026

- **2nd place among 70+ teams, NaSCon '26 Web Dev Hackathon** (team of 2). Multi-tenant CRM that guarantees every inbound lead a human reply: unacknowledged leads escalate down an on-call chain over WhatsApp and Telegram, with live updates in the web app
- Escalation engine runs **inside PostgreSQL** (pg_cron, row locks) for exactly-once hand-offs that survive restarts; row-level-security tenancy, transactional outbox with retries, idempotent signed webhooks (Stripe, Telegram, WhatsApp), and a hash-chained audit log

Identiti: AI Brand Intelligence SaaS | *FastAPI, Supabase, Next.js, Docker* | identiti.studio — Mar 2026 – Present

- **Co-Founder & CTO.** Shipped a live SaaS whose brand-learning engine ingests tone guidelines and past content to generate platform-specific posts for Instagram, LinkedIn, and X, with concurrent image generation via Gemini
- Architected the backend with JWT auth, RBAC, and Dockerized PostgreSQL; supports multi-brand workspaces, A/B variations, and revision history

Portfolio Tracker API | *Node.js, TypeScript, PostgreSQL, Redis, WebSockets, Bull* | GitHub — Feb 2026

- REST API for stock and crypto portfolios; a **Bull queue** drives 5-minute price refreshes decoupled from the request lifecycle, and a **WebSocket server** broadcasts live prices to connected clients
- P&L and CAGR analytics engine; RBAC, rate limiting, and idempotent operations on all critical endpoints; normalized PostgreSQL schema with optimized indexing

RishtaAI: Multi-Agent Matchmaking System | *Gemini, Google Cloud STT, Expo, Node.js, Supabase* — May 2026

- Technical lead of a 5-person team at **Google Hackathon 2026**; architected an **8-agent orchestration** with a persistent negotiation engine, live reasoning traces over SSE, and voice input in Urdu, Roman Urdu, and English

Machine Learning & Computer Vision Projects

BowlSense: Cricket Bowling Analysis from Video | *PyTorch, YOLO-Pose, FastAPI* | GitHub — Aug 2026 – Present

- **Final year project (team of 3).** Estimating a bowler's elbow extension from a single 240 fps smartphone video in place of lab motion capture: fine-tuned 2D pose estimation, 2D-to-3D lifting, event detection, and rejection of clips too poor to measure
- Built the labelling tool and a dataset of **84 deliveries from 14 bowlers**; trained a temporal-convolution ensemble that locates key events **within 3 frames (12.5 ms)** on unseen bowlers; built the FastAPI inference server, Flutter client, and Docker deployment

Lumen: AI Video Understanding & Summarization | *Python, OpenCV, EasyOCR, faster-whisper* | GitHub — May 2026

- Built a **10-stage multimodal pipeline** (speech-to-text, SSIM keyframe extraction, image enhancement, OCR, fusion, LLM summarization) for live streams and recorded lectures; processes a 30-minute video in **232 s on CPU**
- Classical image enhancement (CLAHE, bilateral filtering, Otsu thresholding, deskewing) raised **OCR confidence from 43% to 79%** on whiteboards and from 61% to 84% on slides

GPU-Accelerated KLT Feature Tracker | *CUDA, OpenACC, C/C++* | GitHub — Oct 2025

- **18× speedup** over the CPU baseline with CUDA (coalesced access, pinned memory, shared-memory tiling) and 7× with OpenACC

Probabilistic Urdu Story Generator | *Python, FastAPI, Docker, GitHub Actions* | GitHub — Mar 2026

- **BPE tokenizer from scratch** and a trigram language model trained on 2,143 Urdu stories (∼1.87M words); served with real-time token streaming and CI/CD

Achievements

ICPC Regional 2025 — 21st in Pakistan: Team "Fikr-e-code" (27th overall), sponsored by FAST-NUCES
1st Place, Code Jail (Tech'On '25): Won the university-wide competitive programming challenge with a perfect score
Summer AI Fellow, Headstarter (New York, remote): Selected for a competitive AI fellowship, Jun – Jul 2024
Leadership: Officer, Computing Events, NaSCon '25 (organized the Data Visualization competition); Workshops Team, FAST Data Science Society; volunteer community lecturer in public schools and orphanages
Languages: English (professional), Urdu (native) · **Hafiz-e-Quran**